Mustafa Abdullazade

abdullazadamustafa@gmail.com | +994 50 299 40 52 | Baku, Azerbaijan
abdullazada-mustafa.github.io | github.com/abdullazada-mustafa

Research Interests

My path from mechatronics and robotics to computer science, through a software internship at Nokia and backend development at NIX Tech, has shown me how much modern systems depend on software that is secure and trustworthy. My thesis built a conference management platform hardened against common attacks such as SQL injection and cross-site scripting and load tested it at conference scale. As more software is written by AI models or relies on them to make decisions, I want to understand when those decisions can be trusted. I aim to pursue research on the security and reliability of AI-driven software, including code written by AI.

Education

Budapest University of Technology and Economics (BME) — Budapest, Hungary
M.Sc. Computer Science Engineering — *Sep 2022 – Jan 2026*

- **Thesis:** Conference Management System (supervisor: Kawthar Al-Ghourani)

Azerbaijan State Oil and Industry University (ASOIU) — Baku, Azerbaijan
B.Sc. Mechatronics and Robotics Engineering — *Sep 2018 – Jul 2022*

- Program delivered in cooperation with the University of Siegen, Germany

Master's Thesis

Conference Management System
M.Sc. Thesis, BME, custom platform for a doctoral workshop

- Designed and built a full-stack platform covering the complete peer review lifecycle (abstract submission, paper upload, reviewer assignment, structured scoring, final decisions) with role-based dashboards for authors, reviewers, and administrators.
- Architected a hybrid data layer combining PostgreSQL for transactional data, Google Firestore for real-time status updates, and AWS S3 for papers uploaded directly from the browser via signed URLs.
- Secured the system with JWT authentication, bcrypt password hashing, parameterized SQL queries against injection, input sanitization against cross-site scripting, and atomic transactions.
- Load tested a simulated 2,500-user conference (1,200 papers) with Apache JMeter, k6, and Artillery, keeping 95th-percentile latency under 210 ms at 500 concurrent users with a success rate above 99.8%.
- Reached 3,150 requests per second at peak, profiled PostgreSQL queries to locate CPU and connection pool bottlenecks, and confirmed no memory leaks over a 24-hour stress test.

Work Experience

Mandatory Military Service — Apr 2026 – Apr 2027 (expected)
National Defense University — *Azerbaijan*

Backend Developer — Mar 2024 – Jan 2026
NIX Tech — *Budapest, Hungary*

- Developed and maintained backend modules for core banking operations.
- Designed and optimized Oracle PL/SQL procedures, triggers and packages handling high-volume transactions.
- Built and integrated RESTful APIs connecting internal systems and services.
- Implemented business logic automation for contract and document generation linked to multiple credits and guarantees.
- Managed code versioning and deployment using Git and Team Foundation Server (TFS).
- Participated in internal audits and optimized database logging to improve change traceability.

Software Developer Intern — Jan 2023 – Jan 2024
Nokia — *Budapest, Hungary*

- Engineered backend service modules and implemented RESTful APIs using Python and Java within an Agile development workflow.
- Automated system and integration testing using Python scripts, expanding code coverage and improving build stability.
- Containerized application components using Docker and supported CI/CD pipeline deployments (Jenkins and GitLab).
- Participated in peer code reviews and authored detailed technical documentation for service APIs and system tools.

Honors and Awards

Stipendium Hungaricum Scholarship, Government of Hungary (full M.Sc. scholarship) 2022 – 2026

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL
Web Development: React, Node.js, Express, Vite, Material UI
Data, Cloud, and Tools: PostgreSQL, Oracle PL/SQL, Firebase, AWS S3, Docker, Nginx, Git, Team Foundation Server, Linux
Testing and Monitoring: Apache JMeter, k6, Artillery, pytest, Prometheus, Grafana
Machine Learning: scikit-learn, pandas, LLM APIs (Gemini), local LLMs with Ollama
Languages: Azerbaijani (native), English (fluent), Turkish (fluent)